

SOURCE CODE

ABSTRACT CLASS

```
package model;
```

```
public abstract class User {  
    protected int id;  
    protected String name;  
  
    public User(int id, String name) {  
        this.id = id;  
        this.name = name;  
    }  
  
    public abstract void displayMenu();  
}
```

STUDENT CLASS

```
package model;
```

```
public class Student extends User {  
    private int roomId;  
  
    public Student(int id, String name, int roomId) {  
        super(id, name);  
        this.roomId = roomId;  
    }  
  
    public int getRoomId() { return roomId; }
```

```
public void setRoomId(int roomId) { this.roomId = roomId; }
```

```
@Override
```

```
public void displayMenu() {
```

```
    System.out.println("Student Menu");
```

```
}
```

```
}
```

ADMIN CLASS

```
package model;
```

```
public class Admin extends User {
```

```
    public Admin(int id, String name) {
```

```
        super(id, name);
```

```
}
```

```
@Override
```

```
public void displayMenu() {
```

```
    System.out.println("Admin Menu");
```

```
}
```

```
}
```

ROOM CLASS

```
package model;
```

```
public class Room {
```

```
    private int roomId;
```

```
private int capacity;
```

```
public Room(int roomId, int capacity) {  
    this.roomId = roomId;  
    this.capacity = capacity;  
}
```

```
public int getRoomId() { return roomId; }  
public int getCapacity() { return capacity; }  
}
```

COMPLAINT CLASS

```
package model;
```

```
public class Complaint {  
    private int complaintId;  
    private int studentId;  
    private String status;  
  
    public Complaint(int complaintId, int studentId, String status) {  
        this.complaintId = complaintId;  
        this.studentId = studentId;  
        this.status = status;  
    }  
}
```

DATABASE MANAGER

```
package util;
```

```
import java.sql.Connection;
import java.sql.DriverManager;

public class DatabaseManager {
    private static DatabaseManager instance;
    private Connection connection;

    private DatabaseManager() {
        try {
            connection = DriverManager.getConnection("jdbc:sqlite:hostel.db");
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    public static DatabaseManager getInstance() {
        if (instance == null)
            instance = new DatabaseManager();
        return instance;
    }

    public Connection getConnection() {
        return connection;
    }
}
```

STUDENT DAO

```
package dao;

import model.Student;
import util.DatabaseManager;

import java.sql.Connection;
import java.sql.PreparedStatement;

public class StudentDAO {

    public void addStudent(Student student) {
        try {
            Connection con = DatabaseManager.getInstance().getConnection();
            String query = "INSERT INTO students VALUES (?, ?, ?)";

            PreparedStatement ps = con.prepareStatement(query);
            ps.setInt(1, student.id);
            ps.setString(2, student.name);
            ps.setInt(3, student.getRoomId());

            ps.executeUpdate();
            System.out.println("Student added successfully");

        } catch (Exception e) {
            System.out.println("Error adding student");
        }
    }
}
```

SERVICE LAYER

```
package service;
```

```
import dao.StudentDAO;
```

```
import model.Student;
```

```
public class HostelService {
```

```
    private StudentDAO studentDAO = new StudentDAO();
```

```
    public void registerStudent(int id, String name, int roomId) {
```

```
        Student s = new Student(id, name, roomId);
```

```
        studentDAO.addStudent(s);
```

```
    }
```

```
}
```

MAIN UI

```
package ui;
```

```
import service.HostelService;
```

```
import java.util.Scanner;
```

```
public class MainUI {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        HostelService service = new HostelService();
```

```
while (true) {  
    System.out.println("\n1. Add Student");  
    System.out.println("2. Exit");  
  
    int choice = sc.nextInt();  
  
    switch (choice) {  
        case 1:  
            System.out.print("Enter ID: ");  
            int id = sc.nextInt();  
            sc.nextLine();  
  
            System.out.print("Enter Name: ");  
            String name = sc.nextLine();  
  
            System.out.print("Enter Room ID: ");  
            int roomId = sc.nextInt();  
  
            service.registerStudent(id, name, roomId);  
            break;  
  
        case 2:  
            System.exit(0);  
    }  
}  
}
```